SuddWatch — Phase 1: Preprocessing & Feature Engineering Strategy

04-652 AI System Design · Assignment 2 · Carnegie Mellon University Africa
Jok Akech Atem Mabior · Binary classification · Repository: https://github.com/Super-Talen/SuddWatch

0. Problem, task and data

Objective	Forecast, for each of South Sudan's 79 counties, whether surface-water extent will exceed that county's normal high-water level **13–24 days after the forecast is issued**, so humanitarian planners can pre-position aid before roads and river routes close (SDG 13 → SDG 2, SDG 1).
Task type	**Binary classification** (supervised, short-horizon spatio-temporal panel).
Unit of observation	One county × one 12-day Sentinel-1 window (12 days = Sentinel-1A repeat cycle). The forecast is issued at the end of window *w*.
Target Y (`y_flood_2_4wk`)	1 if the county's Sentinel-1 surface-water fraction in window *w*+2 exceeds the county's own **75th percentile computed on training windows only** (floor: 1 km² of water). A county-specific threshold is required because baseline wetness differs by orders of magnitude between the permanent Sudd swamps and the Equatoria highlands; a single global cut-off would label Sudd counties "flooded" permanently and Equatoria counties never.
Sources (all open)	Copernicus Sentinel-1 GRD (labels + autoregressive features), CHIRPS daily rainfall, SRTM DEM, MERIT Hydro (HAND, upstream area), JRC Global Surface Water — all reduced to county level in Google Earth Engine; OCHA COD-AB county boundaries (HDX); UNOSAT VIIRS water extents 2022–2024 (HDX) used only as independent label validation.
Raw panel	**N = 23,305 rows × P = 26 columns** (20 candidate features + 2 ID columns, 2 datetime columns, 1 split tag, 1 target). Issue dates 2017-01-12 → 2026-08-29.

Label construction (decision log). Sentinel-1 provides radar backscatter, not flood maps. Surface water = per-pixel median VV backscatter in the 12-day window < −18 dB, restricted to terrain within 20 m of the nearest drainage (MERIT HAND), aggregated at 200 m. The −18 dB threshold was **calibrated, not assumed**: among −14…−20 dB it gave the best county-level agreement with UNOSAT's independent VIIRS product (Spearman 0.70 on 2023-09-26).
Validation against UNOSAT (2022–2024, 1,657 county-date pairs) — reported honestly. Across counties agreement is good (pooled Spearman ρ = **0.694**) and the national seasonal cycle agrees moderately (ρ = 0.379 over 27 dates). *Within* counties over time — the comparison that matters for a county-relative target — agreement is uneven: median ρ by state Jonglei 0.51, Central Equatoria 0.39, Lakes 0.18, Unity 0.17, Eastern Equatoria -0.01, Upper Nile -0.03, Western Equatoria -0.14, Warrap -0.46, Western Bahr el Ghazal -0.47; overall median 0.041. It is strongest in Jonglei, the core Sudd state that motivates the project, and weak or negative in drier, more vegetated western/northern counties. Causes: (i) the two products measure different things — SAR detects open water but misses water under dense papyrus/reed canopies, while VIIRS (375 m) also captures partly flooded vegetation; (ii) 2022–2024 is the single-satellite Sentinel-1 period with the sparsest coverage. **Phase 2 actions**: add VH-based flooded-vegetation detection, run a −16 dB sensitivity variant, and report all metrics both nationally and for the Sudd core (Jonglei, Unity, Upper Nile, Lakes).

A. Dataset & Problem Audit

A1. Initial shape

The raw (un-preprocessed) panel `data/raw/suddwatch_panel_unprocessed.csv` has **N = 23,305 observations** (79 counties × 295 issue windows) and **P = 26 columns**, of which **20 are candidate predictive features**.

A2. Variable inventory

Counts by type: Continuous: 17, Datetime: 2, Nominal (ID / join key): 1, Nominal (ID / label): 1, Nominal categorical: 1, Nominal (partition tag, not a feature): 1, Discrete (count): 1, Discrete (cyclical): 1, Binary target (Y): 1. **No ordinal variables exist** in this data: states have no rank, and month is cyclical (December is adjacent to January), so treating it as ordinal would impose a false 12-to-1 distance.

column	type	role	dtype	n_unique	n_missing	pct_missing
adm2_pcode	Nominal (ID / join key)	identifier/meta	str	79	0	0.00
adm2_name	Nominal (ID / label)	identifier/meta	str	79	0	0.00
adm1_name	Nominal categorical	feature	str	11	0	0.00
window_start	Datetime	identifier/meta	datetime64[us]	295	0	0.00
issue_date	Datetime	identifier/meta	datetime64[us]	295	0	0.00
split	Nominal (partition tag, not a feature)	identifier/meta	str	3	0	0.00
n_scenes	Discrete (count)	feature	int64	53	0	0.00
obs_frac	Continuous	feature	float64	2820	0	0.00
water_frac	Continuous	feature	float64	16798	2145	9.20
water_frac_lag1	Continuous	feature	float64	16734	2224	9.54
water_frac_lag2	Continuous	feature	float64	16671	2303	9.88
rain_12d_mm	Continuous	feature	float64	20445	0	0.00
rain_36d_mm	Continuous	feature	float64	21652	0	0.00
rain_90d_mm	Continuous	feature	float64	22962	0	0.00
rain_90d_anom_mm	Continuous	feature	float64	23304	0	0.00
county_area_km2	Continuous	feature	float64	79	0	0.00
elev_mean_m	Continuous	feature	float64	79	0	0.00
elev_std_m	Continuous	feature	float64	79	0	0.00
slope_mean_deg	Continuous	feature	float64	79	0	0.00
hand_mean_m	Continuous	feature	float64	79	0	0.00
hand_lt5m_frac	Continuous	feature	float64	71	0	0.00
twi_mean	Continuous	feature	float64	79	0	0.00
perm_water_frac	Continuous	feature	float64	41	0	0.00
jrc_water_occurrence_pct	Continuous	feature	float64	79	0	0.00
month	Discrete (cyclical)	feature	int64	12	0	0.00
y_flood_2_4wk	Binary target (Y)	target	float64	2	2145	9.20

A3. Baseline missingness (before any processing)

Overall, **1.46%** of all 605,930 cells are missing. **19,708** rows are complete and **3,597** rows (15.4%) contain at least one missing value. Missingness is confined to the Sentinel-1-derived columns:

column	type	n_missing	pct_missing
water_frac	Continuous	2145	9.20
water_frac_lag1	Continuous	2224	9.54
water_frac_lag2	Continuous	2303	9.88
y_flood_2_4wk	Binary target (Y)	2145	9.20

Fig 1.1 - Missingness pattern (red dashes: Sentinel-1B loss, Sentinel-1C/D coverage)

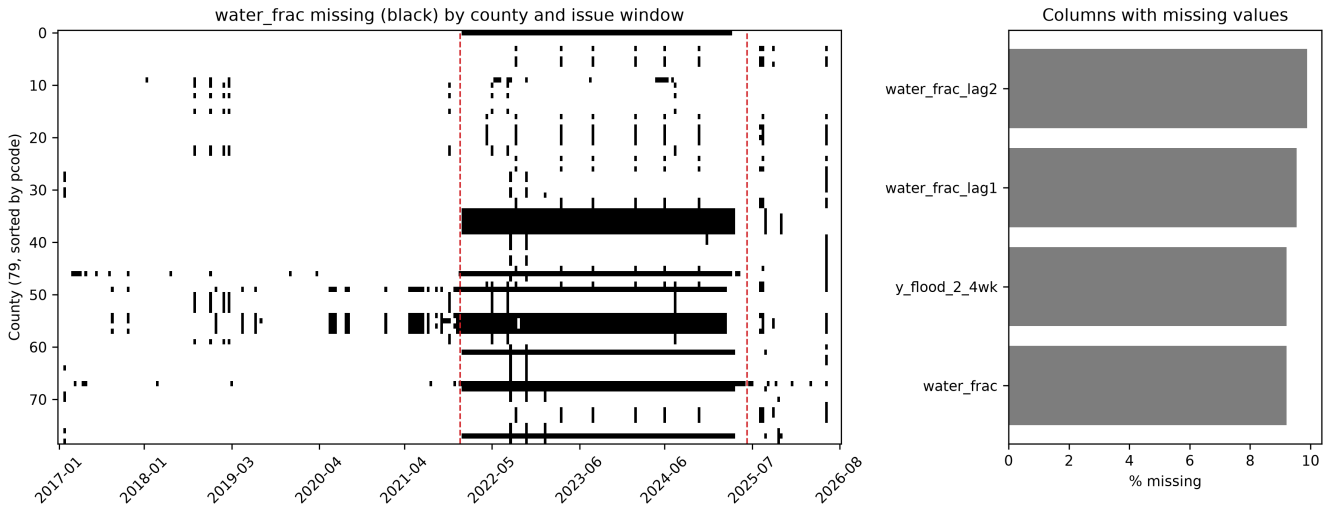


Fig 1.1 — Missingness pattern: water_frac missing by county × issue window (left) and per-column missing % (right).

A4. Target label specification

Binary classification. Class balance of labelled rows per temporal split:

split	issue dates	rows	labelled	Y=1	Y=0	positive rate	imbalance (neg:pos)
train	2017-01-12 → 2023-12-31	16906	15550	2987	12563	19.2%	4.21:1
validation	2024-01-12 → 2025-07-05	3634	2948	1072	1876	36.4%	1.75:1
test	2025-07-17 → 2026-08-29	2765	2662	980	1682	36.8%	1.72:1

Fig 1.2 - Target class balance by temporal split

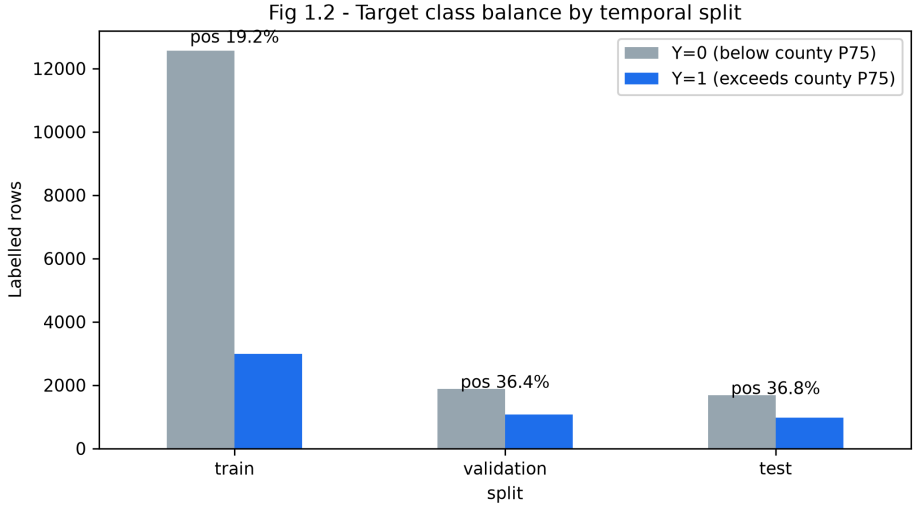


Fig 1.2 — Class balance by split.

Imbalance and mitigation. The training set is moderately imbalanced (4.21:1, 19.2% positive). The rate is below the nominal 25% because of the 1 km² floor: in 24 of 79 dry counties the training P75 is below 1 km² of water, the floor binds, and those counties are positive in only 6.6% of training windows, versus 25.2% in the other 55 counties (≈25% by construction). This is intended: in very dry counties, crossing a sub-1 km² level would reflect radar noise rather than flooding. Positive rates are higher in validation (2024 – mid-2025) and test (mid-2025 – 2026) because the threshold is fixed from 2017–2023 and later seasons were wetter — the 2025 season affected over one million people (proposal, ref. [1]). A sensor effect cannot be fully excluded for the test period (Sentinel-1C/D add scenes from mid-2025); Phase 2 will check it by comparing exceedance rates with and without `n_scenes` stratification. Mitigation plan, in order: (1) `class_weight="balanced"` in the estimator (no synthetic rows, no information loss); (2) choose the decision threshold on the *validation* precision–recall curve to meet the proposal's <15% false-alarm target; (3) evaluate with PR-AUC, F1 and recall rather than accuracy. If resampling (SMOTE) is tried, it runs *inside* training folds only, after the temporal

split, never on validation/test. Random under-sampling is rejected because it discards whole flood seasons from a temporally scarce record.

B. Cleaning & Imputation Strategy

B1. Missingness mechanism classification

Column(s)	Mechanism	Evidence from this dataset	Treatment
`water_frac` (and inherited `obs`-based gaps)	**MAR**	Missing when <50% of the county was imaged in the window. Missing share by satellite era: S1A+S1B (2017-2021): **1.6%**; S1A only (2022-mid 2025): **21.7%**; S1A+S1C/D (mid 2025-2026): **3.7%** — χ^2 test of era × missingness p < 1e-16; by state p < 1e-16 (highest: Northern Bahr el Ghazal 35.5%, Abyei Region 34.6%, Western Bahr el Ghazal 25.8%). The gap is explained by *observed* variables (date → Sentinel-1B failure on 23 Dec 2021 and Sentinel-1C/D arrival in 2025; location → orbit tracks excluded from the single-satellite acquisition plan). **Not MCAR**: it is strongly systematic. **Not MNAR**: ESA plans acquisitions in advance, independently of conditions on the ground. Empirically, in 2022–2024 the raw 36-day rainfall before missing windows is lower (median 77.2 vs 95.3 mm, Mann-Whitney p = 0.005) — but only because the un-imaged counties are drier places: in a logistic regression of missingness on state + month + standardised rainfall, the rainfall coefficient is -0.016 (p = 0.86). Once observed location and season are controlled, wetness carries no information about missingness. (MNAR can never be fully excluded from observed data alone; here it is ruled out by the acquisition mechanism and this adjusted test.)	County-level **training-median** imputation + binary `_was_missing` indicator (see justification below).
`water_frac_lag1`, `water_frac_lag2`	**MAR** (+ structural)	Shifted copies of `water_frac`, so they inherit its MAR gaps; the first 1–2 windows of 2017 are missing by construction (no earlier window exists).	Same imputer + indicator; structural leading gaps fall in early training only.
`y_flood_2_4wk`	**MAR**	Undefined when window *w*+2 was not imaged (same acquisition-plan mechanism).	**Never imputed.** Rows without Y are excluded from fitting and evaluation (imputing a label would fabricate ground truth).
All rainfall, terrain, calendar, ID columns	—	0 missing (gridded products with complete coverage; static county attributes).	None needed.

Why median, not mean. `water_frac` is extremely right-skewed (training skewness = **22.6**; mean = 0.0037 vs median = 0.0004). A mean fill would inject a value inflated by rare floods into ordinary dry windows. **Why per county:** baseline wetness differs by orders of magnitude between counties; a global median would erase that structure. **Why not KNN:** the gaps are spatially clustered — whole orbit tracks are missing together — so a county's nearest neighbours are usually missing in the same window; KNN would borrow from distant, hydrologically unrelated counties. **Why not last-observation-carried-forward:** 9 counties were not imaged for up to ~3 years (2022–2025), so carried-forward values would be years stale. **Mode** is inapplicable to a continuous feature. The indicator columns preserve the information that a value was imputed.

B2. Outlier identification rules (training rows)

Rule chosen by distribution shape: |skew| ≤ 0.5 → Z-score rule (|z| > 3); |skew| > 0.5 → 1.5×IQR rule. Bounds below are computed on training rows only and are re-learned inside the pipeline (`OutlierCapper.fit`) on `X_train`.

feature	skew	mean	median	rule	iqr_low	iqr_high	pct_iqr_outliers	pct_z3_outliers	treatment
obs_frac	-2.90	0.9124	1	IQR (skewed)	0.9993	1	18.7%	5.8%	Not capped: coverage metadata bounded in [0,1]; low values are real acquisition gaps.

feature	skew	mean	median	rule	iqr_low	iqr_high	pct_iqr_outliers	pct_z3_outliers	treatment
water_frac	22.60	0.0037	0.0004	IQR (skewed)	-0.0052	0.0087	12.5%	0.9%	Flag only (IQR) — NOT capped: outliers are flood events (the signal). log1p(100·x) compresses the tail.
water_frac_lag1	22.93	0.0037	0.0004	IQR (skewed)	-0.0052	0.0087	12.5%	0.9%	Same as water_frac.
water_frac_lag2	23.29	0.0037	0.0004	IQR (skewed)	-0.0052	0.0087	12.5%	0.9%	Same as water_frac.
rain_12d_mm	0.50	32.43	29.08	IQR (moderate skew)	-83.24	141	0.0%	0.2%	Cap at training 1.5×IQR fences (skewed).
rain_36d_mm	0.24	97.36	94.38	Z-score (|z|>3)	-233.5	411.7	0.0%	0.0%	Cap at training μ ± 3σ (near-normal).
rain_90d_mm	0.16	242.2	237.4	Z-score (|z|>3)	-465.9	926.7	0.0%	0.0%	Cap at training μ ± 3σ (near-normal).
rain_90d_anom_mm	0.48	7.395	1.527	Z-score (|z|>3)	-69.96	84.07	6.9%	1.2%	Cap at training μ ± 3σ (near-normal).
county_area_km2	4.29	8138	5873	IQR (skewed)	-3421	1.743e+04	6.3%	2.5%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
elev_mean_m	1.91	512.8	432.3	IQR (skewed)	133.7	856.2	6.3%	2.5%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
elev_std_m	3.43	50.81	16.7	IQR (skewed)	-83.02	146.5	7.6%	3.8%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
slope_mean_deg	3.07	0.7605	0.2463	IQR (skewed)	-1.21	2.457	6.3%	3.8%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
hand_mean_m	3.27	6.525	1.493	IQR (skewed)	-13.73	24.08	3.8%	3.8%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
hand_lt5m_frac	-0.70	0.7406	0.9353	IQR (moderate skew)	-0.5856	1.95	0.0%	0.0%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
twi_mean	-0.68	12.24	12.95	IQR (moderate skew)	6.732	17.34	0.0%	0.0%	Static: real geography, not errors → not capped; log1p if |skew| > 1.

feature	skew	mean	median	rule	iqr_low	iqr_high	pct_iqr_outliers	pct_z3_outliers	treatment
perm_water_frac	2.99	0.0009	0	IQR (skewed)	-0.0013	0.0021	15.2%	3.8%	Static: real geography, not errors → not capped; log1p if |skew| > 1.
jrc_water_occurrence_pct	2.34	0.1567	0.0194	IQR (skewed)	-0.3134	0.5238	8.9%	3.8%	Static: real geography, not errors → not capped; log1p if |skew| > 1.

B3. Outlier treatment: capping, not deletion

No row is ever deleted for an outlier. (a) **Preserves N**: deleting every row with any flagged value would remove a large share of the panel and, worse, preferentially remove flood seasons — the minority class. (b) **Prevents information loss**: a row with one extreme rainfall value still carries valid terrain, calendar and water-history signal. (c) **Domain rule**: water-fraction "outliers" are the floods SuddWatch exists to predict, so they are flagged but **not capped**; static terrain extremes are real counties (e.g. the Equatoria highlands), so they are log-transformed, not capped. Only dynamic rainfall features are winsorised, at the training bounds above.

B4a. Feature–target association matrix (training rows)

Matching rule: continuous/discrete feature × binary Y → **point-biserial r** (linear association) **plus Mann-Whitney U** (rank-based, robust to the heavy skew found in B2) **plus mutual information** (captures non-monotonic effects such as seasonality); nominal state × binary Y → **χ^2 test with Cramér's V** and MI. Pearson r is not used between features and Y (Y is binary), and Spearman ρ is used between features (B4b) because most are non-normal. **Caveat:** static terrain features take only 79 distinct values (one per county) repeated over time, so their row-level p-values overstate the evidence; mutual information, not p, is therefore the ranking and tie-break criterion.

feature	type	test	|r_pb| / Cramér's V	p (r_pb / χ^2)	p (Mann-Whitney)	mutual_info
water_frac	Continuous	Point-biserial r + Mann-Whitney U + MI	0.257	<1e-16	<1e-16	0.1188
water_frac_lag1	Continuous	Point-biserial r + Mann-Whitney U + MI	0.239	<1e-16	<1e-16	0.0979
water_frac_lag2	Continuous	Point-biserial r + Mann-Whitney U + MI	0.216	<1e-16	<1e-16	0.0908
rain_36d_mm	Continuous	Point-biserial r + Mann-Whitney U + MI	0.346	<1e-16	<1e-16	0.0779
rain_90d_mm	Continuous	Point-biserial r + Mann-Whitney U + MI	0.340	<1e-16	<1e-16	0.0680
month	Discrete (cyclical)	Point-biserial r + Mann-Whitney U + MI	0.218	<1e-16	<1e-16	0.0623
rain_12d_mm	Continuous	Point-biserial r + Mann-Whitney U + MI	0.311	<1e-16	<1e-16	0.0596
hand_lt5m_frac	Continuous	Point-biserial r + Mann-Whitney U + MI	0.161	<1e-16	<1e-16	0.0382
county_area_km2	Continuous	Point-biserial r + Mann-Whitney U + MI	0.047	<1e-16	<1e-16	0.0381
jrc_water_occurrence_pct	Continuous	Point-biserial r + Mann-Whitney U + MI	0.091	<1e-16	<1e-16	0.0380
slope_mean_deg	Continuous	Point-biserial r + Mann-Whitney U + MI	0.121	<1e-16	<1e-16	0.0379
elev_mean_m	Continuous	Point-biserial r + Mann-Whitney U + MI	0.158	<1e-16	<1e-16	0.0379

feature	type	test	|r_pb| / Cramér's V	p (r_pb / χ^2)	p (Mann-Whitney)	mutual_info
elev_std_m	Continuous	Point-biserial r + Mann-Whitney U + MI	0.091	<1e-16	<1e-16	0.0379
hand_mean_m	Continuous	Point-biserial r + Mann-Whitney U + MI	0.121	<1e-16	<1e-16	0.0379
twi_mean	Continuous	Point-biserial r + Mann-Whitney U + MI	0.160	<1e-16	<1e-16	0.0378
rain_90d_anom_mm	Continuous	Point-biserial r + Mann-Whitney U + MI	0.088	<1e-16	<1e-16	0.0354
obs_frac	Continuous	Point-biserial r + Mann-Whitney U + MI	0.009	2.4e-01	1.3e-03	0.0327
n_scenes	Discrete (count)	Point-biserial r + Mann-Whitney U + MI	0.112	<1e-16	<1e-16	0.0257
adm1_name	Nominal categorical	Chi-square + Cramer's V + MI	0.169	<1e-16		0.0162
perm_water_frac	Continuous	Point-biserial r + Mann-Whitney U + MI	0.071	<1e-16	<1e-16	0.0131

B4b. Multicollinearity protocol and preliminary result

Protocol: (1) pairwise screen with |Spearman ρ| ≥ 0.85 on training rows; (2) decision rule — walk pairs from the strongest |ρ| down and, if both are still present, **drop the feature with the lower mutual information with Y** (keeps the more predictive member); (3) iterate VIF on the survivors, dropping the highest VIF > 10 until all VIF ≤ 10. Pairs found:

feature_a	feature_b	spearman	dropped_by_rule
water_frac	water_frac_lag1	0.944	water_frac_lag1
water_frac	water_frac_lag2	0.926	water_frac_lag2
water_frac_lag1	water_frac_lag2	0.944	water_frac_lag1
rain_12d_mm	rain_36d_mm	0.916	rain_12d_mm
rain_36d_mm	rain_90d_mm	0.866	rain_90d_mm
elev_mean_m	elev_std_m	0.920	elev_std_m
elev_mean_m	slope_mean_deg	0.882	slope_mean_deg
elev_mean_m	hand_mean_m	0.888	hand_mean_m
elev_mean_m	hand_lt5m_frac	-0.883	elev_mean_m
elev_mean_m	twi_mean	-0.882	twi_mean
elev_std_m	slope_mean_deg	0.894	elev_std_m
elev_std_m	hand_mean_m	0.908	hand_mean_m
elev_std_m	hand_lt5m_frac	-0.851	elev_std_m
elev_std_m	twi_mean	-0.863	twi_mean
slope_mean_deg	hand_mean_m	0.960	hand_mean_m
slope_mean_deg	hand_lt5m_frac	-0.907	slope_mean_deg
slope_mean_deg	twi_mean	-0.950	twi_mean
hand_mean_m	hand_lt5m_frac	-0.950	hand_mean_m
hand_mean_m	twi_mean	-0.978	twi_mean
hand_lt5m_frac	twi_mean	0.964	twi_mean
perm_water_frac	jrc_water_occurrence_pct	0.894	perm_water_frac

Resulting drop list (applied in Phase 2 via `DROP_COLLINEAR` in `src/pipeline_builder.py`):

feature	reason
twi_mean	|rho|=0.978 with hand_mean_m

feature	reason
hand_mean_m	|rho|=0.960 with slope_mean_deg
water_frac_lag1	|rho|=0.944 with water_frac
water_frac_lag2	|rho|=0.926 with water_frac
elev_std_m	|rho|=0.920 with elev_mean_m
rain_12d_mm	|rho|=0.916 with rain_36d_mm
slope_mean_deg	|rho|=0.907 with hand_lt5m_frac
perm_water_frac	|rho|=0.894 with jrc_water_occurrence_pct
elev_mean_m	|rho|=0.883 with hand_lt5m_frac
rain_90d_mm	|rho|=0.866 with rain_36d_mm

feature	VIF_before	VIF_after
slope_mean_deg	270.75	
hand_mean_m	225.65	
twi_mean	202.42	
hand_lt5m_frac	174.91	1.28
elev_std_m	19.51	
rain_36d_mm	12.52	1.10
elev_mean_m	11.60	
jrc_water_occurrence_pct	10.66	1.26
perm_water_frac	9.63	
rain_12d_mm	6.06	
rain_90d_mm	4.92	
water_frac_lag1	1.81	
water_frac	1.77	1.10
water_frac_lag2	1.75	
county_area_km2	1.42	1.04
rain_90d_anom_mm	1.16	1.11
n_scenes	1.15	1.14
obs_frac	1.11	1.10

Retained numeric features: `obs_frac`, `water_frac`, `rain_36d_mm`, `rain_90d_anom_mm`, `county_area_km2`, `hand_lt5m_frac`, `jrc_water_occurrence_pct`, `n_scenes`. The terrain block (elevation, slope, HAND, TWI) is one physical gradient (highland → floodplain), so one representative suffices. The autoregressive signal is retained through `water_frac` (the latest observed water state, two windows before the target); Phase 2 will test a less collinear momentum feature (`water_frac - water_frac_lag1`) in place of raw lags.

C. Encoding, Scaling & Transformation Strategy

C1. Encoding matched to cardinality

Variable	Type / cardinality	Encoding	Justification
`adm1_name` (state)	Nominal, 11 levels	One-hot (`handle_unknown="ignore"`)	Low cardinality, no order; 11 binary columns is cheap and keeps states distinct.
`adm2_pcode` (county)	Nominal, 79 levels	**Not encoded** — join key, imputation group and CV group only	79 one-hot columns on ~295 rows per county would let a model memorise county identity; county behaviour enters through terrain and water history instead.
`month`	Discrete, cyclical 1–12	sin/cos pair	Preserves Dec–Jan adjacency; an ordinal 1–12 code would place them 11 units apart.
Ordinal variables	None present	OrdinalEncoder not applicable	No variable carries an inherent rank; forcing one would invent order.

C1a. Numerical scaling matched to distributions and algorithms

Feature group	Distribution (B2)	Transform → scaler	Reason
Water fractions	Extreme right skew (≈23)	`log1p(100·x)` → StandardScaler	Log compresses the flood tail; MinMax alone would squash 99% of values near 0.
County area, elevation mean/std, slope, HAND, water occurrence	|skew| > 1	`log1p` → StandardScaler	Multiplicative scales; log makes them closer to symmetric before standardisation.
Rainfall totals and anomaly, TWI, scene count	|skew| ≤ 1	StandardScaler (after capping)	Near-symmetric; mean 0 / sd 1 required by the logistic-regression baseline (L2 penalty) and by PCA (Fig 5.1).
`obs_frac`, `hand_lt5m_frac`	Bounded [0, 1]	MinMaxScaler	Known physical bounds; preserves exact 0 and 1.

The primary model (gradient-boosted trees) is invariant to monotone scaling, so scaling is not needed for it; it is still applied for the logistic-regression baseline, PCA and any distance-based comparison. `build_preprocessor(scale_tree_inputs=False)` skips it for trees.

C2. Train / validation / test partitioning and leakage prevention

(a) Ratios. Temporal blocks of whole flood seasons: train = issue dates < 2024-01-01, validation = 2024-01-01 – 2025-06-30, test = 2025-07-01 – 2026-08-31. Of 21,160 labelled rows this gives **73.5% / 13.9% / 12.6%**. Seven training seasons (2017–2023) cover wet and dry years; the held-out test year contains the full 2025 flood season named in the proposal. With N ≈ 23k an 80/20 random split would be statistically ample, but randomness is the wrong criterion for a forecasting task (see d).

(b) Stratification. `StratifiedShuffleSplit` is **rejected**: it mixes future windows into training (look-ahead leakage) and, because neighbouring windows are autocorrelated, inflates scores. Class proportions are instead *reported* per split (A4) and per CV fold, and imbalance is handled by class weights and validation-tuned thresholds.

(c) Grouping. Every county appears in every split *by design*: the deployed system forecasts the same 79 counties forward in time, so the operational question is temporal generalisation. A supplementary `GroupKFold` by county (within the training period) will measure spatial generalisation to unseen counties as a robustness check.

(d) Temporal ordering. All splits move forward in time. Inside training, hyper-parameters are tuned with `TimeSeriesSplit(n_splits=5, gap=2 windows × 79 rows)`: the 2-window gap purges rows whose target window (w+2) would overlap the next validation fold. The last two training windows are likewise purged before validation.

(e) Leakage-prevention architecture.

Leakage path	Control
Target threshold sees the future	County P75 computed on training windows only (`data/raw/county_thresholds_train_only.csv`).
Rainfall from after the issue date	Sums end on the issue date; anomaly climatology from 2001–2016, strictly before the study period.
Future satellite data	Only windows w, w−1, w−2 enter X; window w+1 and the target window w+2 never do.
Imputation / capping / scaling / encoding statistics	All live in one scikit-learn `Pipeline` (`src/pipeline_builder.py`): `GroupMedianImputer → OutlierCapper(Z) → OutlierCapper(IQR) → ColumnTransformer → estimator`. `fit` is called on `X_train` only and re-fitted inside every CV fold; validation/test are only `transform`ed.
Feature selection	Collinearity and MI statistics computed on training rows only.
Same-time proxies of Y	Excluded: no same-window flood/population layers (e.g. UNOSAT "Pop exposed"); UNOSAT is used only for label validation.

Reproducibility

All code is in `src/`: `gee_extract.py` (Earth Engine extraction, cached and resumable) → `build_panel.py` (panel and target) → `audit.py` (every number and Figs 1.1–1.2) → `report_phase1.py` (this report). Environment: `requirements.txt`. Audit tables: `phase1_strategy/audit/`.